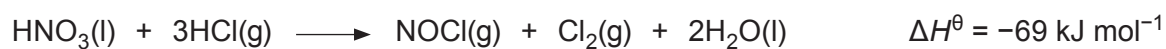


10. Aqua regia is an acidic liquid that can dissolve unreactive metals such as gold. It can be formed by bubbling hydrogen chloride gas through concentrated nitric acid.



Substance	Standard enthalpy change of formation, $\Delta_f H^\theta / \text{kJ mol}^{-1}$	Standard entropy, $S^\theta / \text{J K}^{-1} \text{mol}^{-1}$
$\text{HNO}_3(\text{l})$	-173	156
$\text{HCl}(\text{g})$		187
$\text{NOCl}(\text{g})$	53	264
$\text{Cl}_2(\text{g})$	0	223
$\text{H}_2\text{O}(\text{l})$	-286	70

- (a) Show that this is a redox reaction. [1]

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- (b) State why the enthalpy change of formation of $\text{Cl}_2(\text{g})$ is zero. [1]

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- (c) Calculate the standard enthalpy change of formation of $\text{HCl}(\text{g})$. [2]

$$\Delta_f H^\theta = \dots\dots\dots \text{kJ mol}^{-1}$$



- (d) (i) Calculate the free energy change, $\Delta G^\ominus$, for this reaction at 25 °C. [3]

$$\Delta G^\ominus = \dots\dots\dots \text{kJ mol}^{-1}$$

- (ii) An alternative method of preparing aqua regia is to use concentrated hydrochloric acid with concentrated nitric acid. State, giving a reason, the effect of this on the value of $\Delta G^\ominus$ at 25 °C. [1]

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- (e) (i) The pH of a dilute solution of nitric acid is measured using a pH probe and has a value of 0.3. Calculate the concentration of this solution. [1]

$$\text{Concentration} = \dots\dots\dots \text{mol dm}^{-3}$$

- (ii) An alternative way of measuring the concentration of the solution is by titration against a standard solution of sodium hydrogencarbonate of concentration 0.500 mol dm⁻³.

State and explain which of these two methods will give the more precise value for the concentration. [2]

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